✅ 1. Gumbel model

$$f(x) = \frac{1}{\alpha} \exp\left(-\frac{x-A}{\alpha}\right) \exp\left(-e^{-\frac{x-A}{\alpha}}\right), \quad F(x) = \exp\left(-e^{-\frac{x-A}{\alpha}}\right)$$

Define:

$$y_{n,t} = x_{l_n} - \theta t, \qquad u_{n,t} = \frac{y_{n,t} - A}{\alpha}, \qquad e_{n,t} = e^{-u_{n,t}}$$

✅ 2. Log-likelihood

From your likelihood structure:

$$\boxed{\begin{aligned} \ell(\theta, A, \alpha) = -N_T \log \alpha - \sum_{n=1}^{N_T} \frac{y_{n,l_n} - A}{\alpha} \\ - \sum_{n=1}^{N_T} \sum_{t \in \mathcal{I}_n} \exp\left(-\frac{y_{n,t} - A}{\alpha}\right) \end{aligned}}$$

where:

$$\mathcal{I}_n = \begin{cases} \{l_n, \ldots, l_{n+1} - 1\}, & n < N_T \\ \{l_{N_T}, \ldots, T\}, & n = N_T \end{cases}$$

✅ 3. First derivatives (summary)

θ

$$\frac{\partial \ell}{\partial \theta} = \frac{1}{\alpha} \sum_n l_n - \frac{1}{\alpha} \sum_{n,t} t\, e_{n,t}$$

A

$$\frac{\partial \ell}{\partial A} = \frac{N_T}{\alpha} - \frac{1}{\alpha} \sum_{n,t} e_{n,t}$$

α

$$\frac{\partial \ell}{\partial \alpha} = -\frac{N_T}{\alpha} + \frac{1}{\alpha^2} \sum_n (y_{n,l_n} - A) - \frac{1}{\alpha^2} \sum_{n,t} (y_{n,t} - A) e_{n,t}$$

✅ 4. Hessian components (used in code)

We compute sums over all (n, t) pairs:

- $S_e = \sum e_{n,t}$
- $S_{te} = \sum t e_{n,t}$
- $S_{t^2e} = \sum t^2 e_{n,t}$
- $S_{ye} = \sum (y_{n,t} - A) e_{n,t}$
- $S_{tye} = \sum t (y_{n,t} - A) e_{n,t}$
- $S_{y^2e} = \sum (y_{n,t} - A)^2 e_{n,t}$